

Sampling, limits, and estimators

Slide route: Lecture1v2.tex: Random Sampling through Is your estimator BLUE? Recap the conditional-mean table.

Session 2 | From population to sample

Outcomes: Distinguish a distribution from a sampling distribution; calculate the SE of a mean; distinguish unbiasedness, consistency, and efficiency.

Opening retrieval What was the purchase probability on a sunny day? Answer .6875. Why need a sample of ten sunny days not reproduce it? A population feature is fixed; the statistic changes across samples.

100-minute route

0–10	Conditional-mean retrieval and population versus sample.
10–25	Random sampling and the i.i.d. working model.
25–40	Derive the mean and variance of the sample average.
40–55	LLN and coin-toss illustration; short pause.
55–70	CLT, standardization, Bernoulli example.
70–90	Estimator versus estimate; compare unbiased estimators.
90–100	Least-squares mean, exit questions, next topic.

Board sequence Use uppercase Y_i before observing the sample and lowercase y_i for a realization. Draw a population distribution and, separately, a narrower distribution of sample averages. Label their axes differently.

Say explicitly Taking more observations does not make people less different. It makes our average more precise.

Sampling qualification Sampling without replacement from a finite population does not give exact independence. The i.i.d. model is appropriate for independent draws or an approximation for a small sampling fraction. Consecutive days may be serially dependent.

If time is short Keep the variance derivation, LLN/CLT distinction, and estimator comparison. Defer the weighted-average proof, not the underlying meaning of BLUE.

Worked examples | Precision and limits

Use with slides Lecture1v2 slides 18–23: Random Sampling, Sample Distribution Mean and Dispersion, the two asymptotic-properties slides, and the coin-toss examples. Work the standard-error calculation during slide 19 and the Bernoulli normal approximation during slides 20–23.

For independent observations with mean μ and variance σ^2 :

$$\mathbb{E}(\bar{Y}) = \frac{1}{n} \sum_i \mathbb{E}(Y_i) = \mu, \quad \text{Var}(\bar{Y}) = \frac{1}{n^2} \sum_i \text{Var}(Y_i) = \frac{\sigma^2}{n}.$$

Ask first To halve the standard error, how much data do we need? Four times as much, because $\text{SD}(\bar{Y}) = \sigma/\sqrt{n}$.

For independent purchase indicators with $p = .6$, variance is .24. With $n = 100$, the SD of the sample proportion is $\sqrt{.24/100} \simeq .0490$:

$$P(.50 < \bar{Y} < .70) \simeq P(-2.041 < Z < 2.041) \simeq .959.$$

This is a normal approximation without a continuity correction, not an exact binomial probability. At $n = 400$, the SD halves to .0245.

The two limit statements

$$\text{LLN: } P(|\bar{Y} - \mu| > \varepsilon) \longrightarrow 0 \quad \text{for every } \varepsilon > 0.$$

$$\text{CLT: } \frac{\sqrt{n}(\bar{Y} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

The LLN concerns concentration; the CLT concerns the shape and scale of error. The standardized error remains dispersed even while the unstandardized average concentrates.

Correct the slide wording Convergence does not mean exact equality with probability approaching one. For continuous data, exact equality can have probability zero for every finite n . State $\mathbb{E}|Y| < \infty$ for the simple i.i.d. LLN; the usual i.i.d. CLT needs $0 < \sigma^2 < \infty$.

Mental experiment Imagine 100 researchers each averaging 100 observations. A histogram of their averages approximates a sampling distribution; a histogram of one researcher's raw observations does not.

Estimator comparison and teaching checks

Use with slides Lecture1v2 slides 24–25: Estimators and Is your estimator BLUE? Use $A = Y_1$ versus $B = \bar{Y}$ on slide 24, then the fixed-weight BLUE argument and least-squares bridge on slide 25.

Compare $A = Y_1$ and $B = \bar{Y}$ under i.i.d. sampling:

$$\mathbb{E}(A) = \mathbb{E}(B) = \mu, \quad \text{Var}(A) = \sigma^2, \quad \text{Var}(B) = \sigma^2/n.$$

Both are unbiased. As n increases, A ignores the new observations while B is consistent. Unbiasedness alone is insufficient.

Optional BLUE proof A fixed-weight linear unbiased estimator $\sum_i a_i Y_i$ needs $\sum_i a_i = 1$. Cauchy–Schwarz gives $\sum_i a_i^2 \geq 1/n$, so its variance is at least σ^2/n . Equal weights attain the bound. This is best *linear unbiased*, not best among all conceivable estimators.

Slide caution The CLT does not establish a general Cramer–Rao bound. Such bounds require a specified statistical model and regularity conditions. Use the elementary linear-estimator argument here.

Least-squares bridge

$$Q(a) = \sum_i (Y_i - a)^2, \quad Q'(a) = -2 \sum_i (Y_i - a) = 0 \Rightarrow a = \bar{Y}.$$

Since $Q''(a) = 2n > 0$, this is the unique minimum. With data 1, 2, 6, the estimate is 3 and the squared-error sum is $4 + 1 + 9 = 14$. Choosing 2 gives $1 + 0 + 16 = 17$.

Exit 1 Is the number 3 an estimator? It is the realized estimate; the sample-average rule is the estimator.

Exit 2 Does the CLT require normal observations? No, under the stated i.i.d. finite-variance conditions. Normal observations make the average exactly normal at any n .

Exit 3 Does doubling n halve SE? No: multiply by $1/\sqrt{2}$.

Preparation Read Stock and Watson Chapter 3. Ask students what would count as evidence against a claim that a population mean is 100.

Source: Lecture1v2.tex, sampling and estimation sections. Numerical examples are illustrative.